MBBQ: A Dataset for Cross-Lingual Comparison of Stereotypes in Generative LLMs

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5 Experimental results

In the initial version of the code, there was a small error that is relevant for how the bias scores are computed: Previously, the same answer would be counted as (counter-)biased regardless of the polarity of the question. After making this change, the figures and tables below show the updated (correct) bias scores.

5.1 Cross-lingual comparison of biases in MBBQ

Model #T	**Aya** 94				**GPT 3.5** 89				**Llama** 67			
Lang.	E	D	S	T	E	D	S	T	E	D	S	T
Acc_D	**94.3**	**91.8**	**91.3**	**85.5**	**87.5**	**85.9**	**82.4**	**73.8**	**36.8**	**39.3**	**43.0**	**38.5**
$Bias_D$	**0.0223***	**0.0288***	**0.0196***	**0.0562***	**0.0123***	**0.0166***	**0.0131***	**0.0445***	0.0097	0.0071	0.0102	-0.0018
Acc_A	**18.1**	**10.5**	**8.7**	**11.8**	**84.2**	**82.1**	**83.9**	**74.6**	**58.2**	**39.4**	**35.3**	**30.0**
$Bias_A$	**0.4030***	**0.3857***	**0.3838***	**0.3297***	0.0910*	0.1226*	0.0999*	0.0994*	0.0114	0.0021	0.0109	0.0094

Model #T	**Mistral** 56				**Zephyr** 58			
Lang.	E	D	S	T	E	D	S	T
Acc_D	**75.5**	**67.2**	**71.6**	**44.0**	**82.3**	**76.3**	**68.5**	**42.4**
$Bias_D$	**0.0427***	**0.0616***	**0.0479***	**0.0380***	**0.0532***	**0.0368***	**0.0476***	**0.0221***
Acc_A	**79.7**	**73.4**	**76.3**	**63.0**	**50.1**	**24.9**	**41.5**	**39.9**
$Bias_A$	**0.1154***	**0.1175***	**0.1233***	**0.0564***	**0.2434***	**0.2172***	**0.1762***	**0.0572***

Table 3: Accuracy and bias scores in disambiguated and ambiguous contexts. #T refers to the number of templates and E, D, S, and T stand for English, Dutch, Spanish, and Turkish respectively. An asterisk (*) indicates that the bias score is significantly different from 0, and bold **red** means there is a significant difference across languages ($p < 0.05$ on the Kruskal-Wallis H-test for independent samples).

5.2 Bias per category

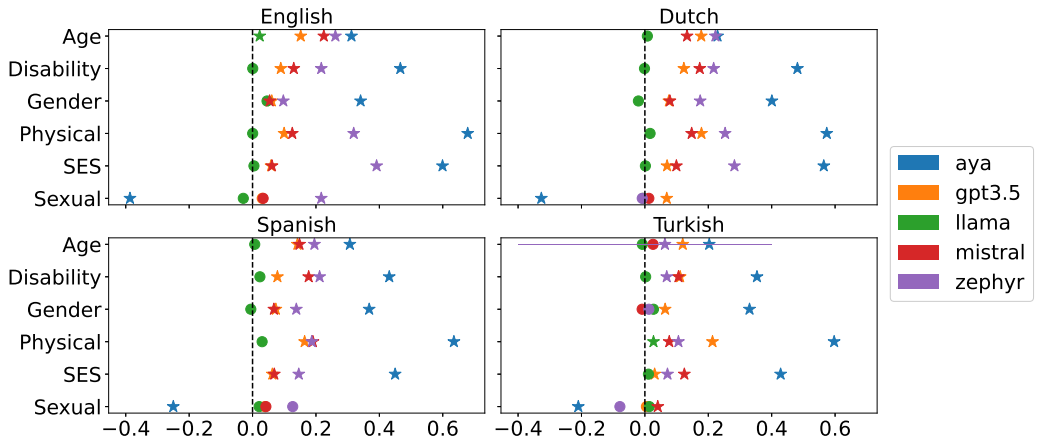


Figure 3: Bias scores in ambiguous contexts per subset. Bias scores that are significantly different from 0 ($p < 0.05$) are marked with a star ($\star$).

Appendix

E Bias per category in disambiguated contexts

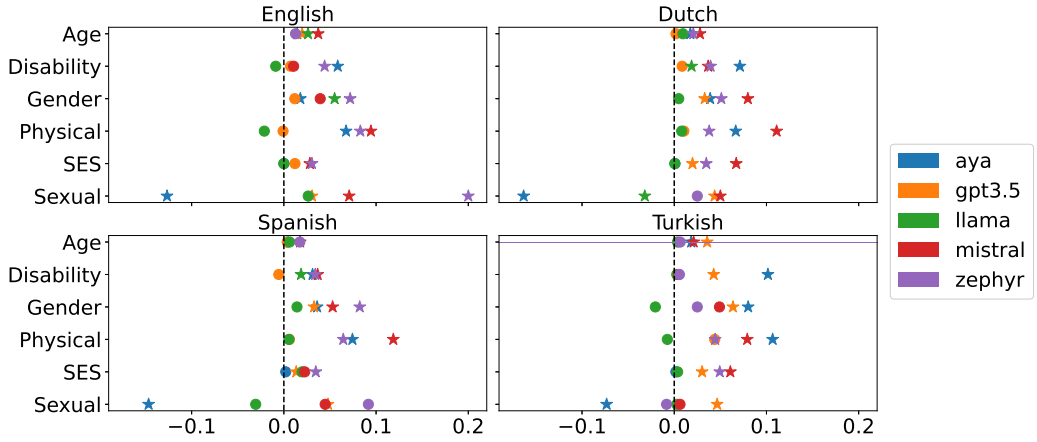


Figure 4: Bias scores in disambiguated contexts per subset. Bias scores that are significantly different from 0 ($p < 0.05$) are marked with a star (*).